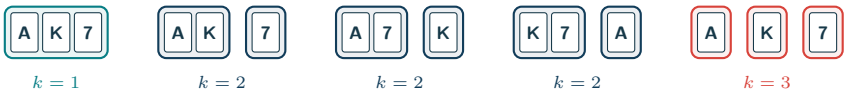
COMBINATORICS ON ONE PAGE

Cards, piles and finite differences

The x^n deals of n cards into x piles are Newton's difference expansion with Worpitzky coefficients.

1 / THE RULE AND THE PICTURE

Partition the cards into heaps, hand the heaps to piles, and sum over the number of piles used



$k=1$ $k=2$ $k=2$ $k=2$ $k=3$

the five partitions of three cards, $B_3 = 5$, with $S(3,k) = 1, 3, 1$ for $k = 1, 2, 3$

Deal n distinct cards into x labelled piles, any pile allowed to stay empty, in x^n ways. If exactly k piles are used, choose them in $\binom{x}{k}$ ways, partition the cards into k heaps in $S(n,k)$ ways and hand the heaps to the chosen piles in $k!$ ways:

$$x^n = \sum_k \binom{x}{k} k!\, S(n,k) = \sum_k \binom{x}{k} W(n,k)$$

2 / A COUNT TO CHECK / THREE CARDS, THREE PILES

The twenty-seven deals split as $3 + 18 + 6$ by the number of piles used

k	$S(3,k)$	$\times\, k! = W(3,k)$	$\binom{3}{k}$ piles	deals	example
1	1	1	3	3	all three cards on one pile
2	3	6	3	18	two cards together, one apart
3	1	6	1	6	one card on each pile
total	$B_3 = 5$			$27 = 3^3$	

3 / FINITE DIFFERENCES / THE SAME NUMBERS ON THE LEFT EDGE

The kth difference of x^n at 0 is $W(n,k)$, and a difference is inclusion and exclusion

x^4	0	1	16	81	256	625
Δ	1	15	65	175	369	
Δ^2	14	50	110	194		
Δ^3	36	60	84			
Δ^4	24	24				
Δ^5	0					

The values of x^4 at $x = 0, \ldots, 5$ and their successive differences have the left edge $0, 1, 14, 36, 24$, which is row 4 of the Worpitzky numbers, and the fourth difference is the constant $24 = 4!$. The formula

$$\Delta^k x^n \big|_0 = \sum_j (-1)^{k-j} \binom{k}{j} j^n = W(n,k)$$

counts the deals onto j piles, removes those that leave a pile empty, and adds back the overlaps, so each difference removes the empty piles. Newton's expansion $p(x) = \sum_k \Delta^k p(0) \binom{x}{k}$ then reproduces $x^n = \sum_k W(n,k) \binom{x}{k}$.

4 / THE TWO EXTREMES / COARSEST AND FINEST PARTITION

The first and last non-zero differences are all cards in one pile and all cards alone

For $k = 1$ every card lies on the same pile and $W(n,1) = 1$, the coarsest partition, while for $k = n$ every card is alone and the n piles are ordered, giving $W(n,n) = n!$, the finest. The next difference is zero because n cards cannot occupy $n+1$ piles, which is why a polynomial of degree n has a constant nth difference $n!$ times its leading coefficient and nothing beyond.

5 / A POKER VERSION / RANK SEQUENCES OF FIVE CARDS

Ignoring suits, $13^5 = \sum_k \binom{13}{k} W(5,k)$ with k the number of distinct ranks

k	$S(5,k)$	$W(5,k)$	$\binom{13}{k}$	ordered rank sequences	hand patterns
1	1	1	13	13	(5), impossible with four suits
2	15	30	78	2 340	$(4,1)$ four of a kind; $(3,2)$ full house
3	25	150	286	42 900	$(3,1,1)$ three of a kind; $(2,2,1)$ two pair
4	10	240	715	171 600	$(2,1,1,1)$ one pair
5	1	120	1287	154 440	$(1,1,1,1,1)$ no pair, straight, flush

The terms sum to $371\,293 = 13^5$, and the hand patterns are the integer partitions of 5 that refine each number of distinct ranks. With four suits the pattern (5) cannot occur, so real hand counts need that cap.

ASIDE / BELL AND THE PACK

The partitions of the five positions of a hand into equal-rank groups number $B_5 = 52$, the size of the pack, which looks like a coincidence.

LINKS TO OTHER SHEETS

The triangles and their gauge: *The general sequencing law for triangles*; hand counts with four suits: the poker hands sheet.

Convention. $W(n,k) = k!\, S(n,k)$, S the Stirling numbers of the second kind, $\Delta f(x) = f(x+1) - f(x)$. The cards are distinct and the piles are labelled.

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